

RentProjectWIP

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```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2     4.0.2      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr       1.2.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(readxl)
```

```
library(dplyr)
```

```
library(ggplot2)
```

```
library(tidyr)
```

```
library(readr)
```

```
RentWIP <- read_csv("Y:/DATA222/Data Sets/apartments_for_rent_classified_10K.csv")
```

Warning: One or more parsing issues, call `problems()` on your data frame for details,
e.g.:

```
dat <- vroom(...)
```

```
problems(dat)
```

Rows: 10000 Columns: 1

```
-- Column specification -----
```

Delimiter: ","

chr (1): id;category;title;body;amenities;bathrooms;bedrooms;currency;fee;ha...

i Use `spec()` to retrieve the full column specification for this data.
 i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
RentWIP <- read_delim(
  "Y:/DATA222/Data Sets/apartments_for_rent_classified_10K.csv",
  delim = ";"
)
```

Rows: 10000 Columns: 22

-- Column specification -----

Delimiter: ";"

chr (18): category, title, body, amenities, bathrooms, bedrooms, currency, f...

dbl (4): id, price, square_feet, time

i Use `spec()` to retrieve the full column specification for this data.
 i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
RentWIP <- RentWIP %>%
  rename_with(tolower) #converts entries into all lowercase
head(RentWIP)
```

A tibble: 6 x 22

	id	category	title	body	amenities	bathrooms	bedrooms	currency	fee
	<dbl>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	5668626895	housing/re~	Stud~	This~	null	null	0	USD	No
2	5664597177	housing/re~	Stud~	This~	null	null	1	USD	No
3	5668626833	housing/re~	Stud~	This~	null	1	0	USD	No
4	5659918074	housing/re~	Stud~	This~	null	1	0	USD	No
5	5668626759	housing/re~	Stud~	This~	null	null	0	USD	No
6	5667891676	housing/re~	0 BR~	**RA~	Dishwash~	1	0	USD	No

i 13 more variables: has_photo <chr>, pets_allowed <chr>, price <dbl>,
 # price_display <chr>, price_type <chr>, square_feet <dbl>, address <chr>,
 # cityname <chr>, state <chr>, latitude <chr>, longitude <chr>, source <chr>,
 # time <dbl>

```
RentWIP <- RentWIP %>%
  mutate(
    price = as.numeric(price),
    square_feet = as.numeric(square_feet),
    bathrooms = as.numeric(bathrooms),
```

```
bedrooms = suppressWarnings(as.numeric(bedrooms))
) #converts these columns into numerical values
```

Warning: There was 1 warning in `mutate()`.
 i In argument: `bathrooms = as.numeric(bathrooms)`.
 Caused by warning:
 ! NAs introduced by coercion

```
RentWIP <- RentWIP %>%
  mutate(
    bathrooms = if_else(is.na(bathrooms), 0, bathrooms)
  )
```

```
RentWIP <- RentWIP %>%
  mutate(
    title = str_squish(str_to_sentence(title)), #lowercases titles, removes extra spaces
    body = str_squish(body), #removes extra spaces
    amenities = str_squish(str_to_lower(amenities)) #makes everything lowercase and removes
  )
RentWIP %>%
  count(pets_allowed)
```

```
# A tibble: 5 x 2
  pets_allowed      n
  <chr>          <int>
1 Cats           485
2 Cats,Dogs     5228
3 Dogs           124
4 None          2415
5 null          1748
```

```
RentWIP <- RentWIP %>%
  mutate(across(where(is.character), str_to_lower)) %>% #converts character columns into lower
  mutate(
    pets_allowed = as.factor(pets_allowed), #turns the following columns into factors
    state = as.factor(state),
    cityname = as.factor(cityname),
    fee = as.factor(fee)
  )
```

```
RentWIP <- RentWIP %>%
  mutate(
    pets_allowed = case_when(
      pets_allowed == "cats,dogs" ~ "cats,dogs",
      pets_allowed == "cats" ~ "cats",
      pets_allowed == "dogs" ~ "dogs",
      pets_allowed == "none" ~ "none",
      TRUE ~ "none"
    )
  ) #makes sure empty entries for pets_allowed are converted to 'none'
```

```
RentWIP <- RentWIP %>%
  mutate(
    amenities = case_when(
      amenities == "null" ~ NA_character_,
      TRUE ~ amenities
    )
  ) #converts null entries in amenities column into a N/A character
```

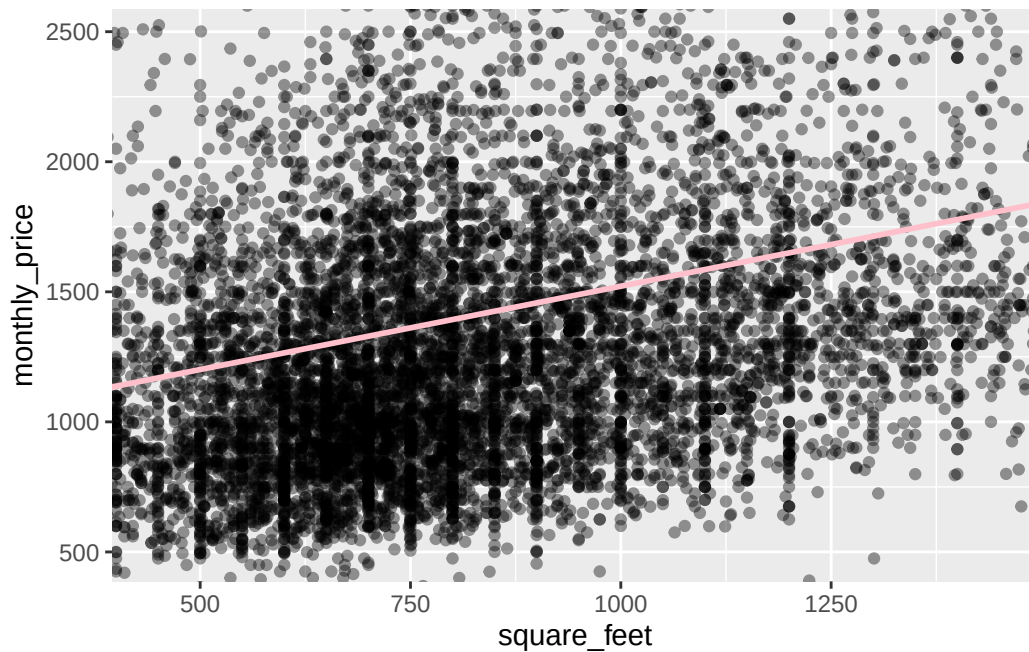
```
unique(RentWIP$price_type)
```

```
[1] "monthly"      "weekly"      "monthly|weekly"
```

```
RentWIP <- RentWIP %>%
  mutate(
    monthly_price = case_when(
      price_type == "monthly" ~ price,
      price_type == "weekly" ~ price * 4.33,
      price_type == "monthly|weekly" ~ price,
      TRUE ~ NA_real_
    )
  ) #calculates and generates a monthly price column so that all listings are priced at a mo
```

```
mean_sqft <- mean(RentWIP$square_feet, na.rm = TRUE) #calc avg monthly rent and square foot
mean_price <- mean(RentWIP$monthly_price, na.rm = TRUE)
ggplot(RentWIP, aes(x = square_feet, y = monthly_price)) +
  geom_point(alpha = 0.4) + #gives each point on scatter plot 40% opacity
  geom_smooth(method = "lm", se = FALSE, color = "pink") + #can see a positive linear relation
  coord_cartesian(
    xlim = c(mean_sqft - 500, mean_sqft + 500),
    ylim = c(mean_price - 1000, mean_price + 1000)
  ) #adds a range of focus without dropping out outliers, focused on a range around the avg s
```

```
`geom_smooth()` using formula = 'y ~ x'
```

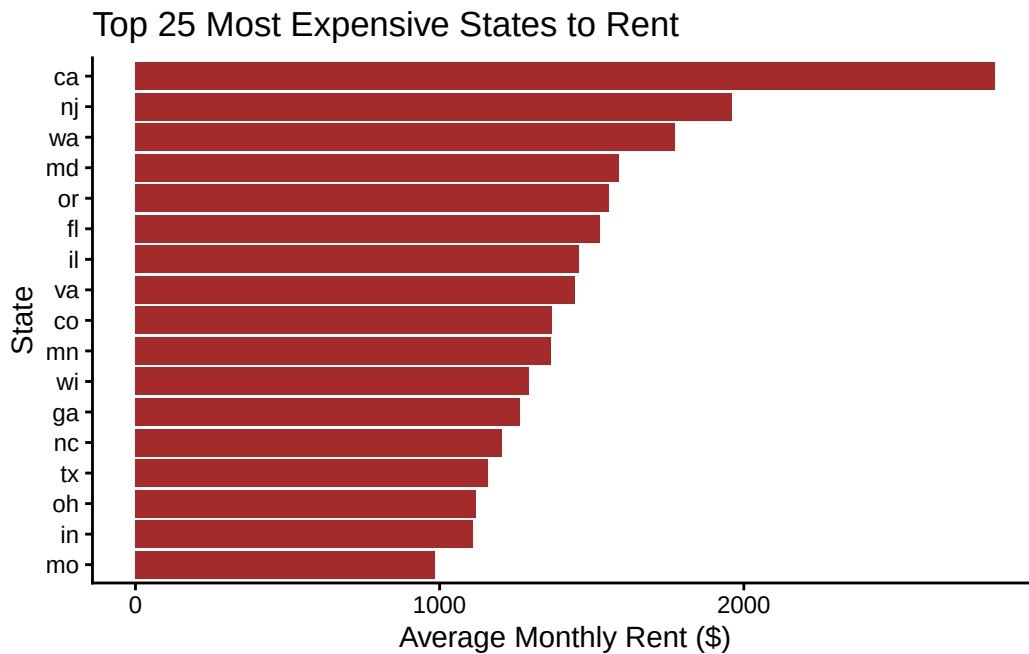


```
state_count <- RentWIP %>%  
  filter(state != "null") %>% #filters out any null entries for the state column  
  count(state) #counts how many listings for each state  
avg_count <- mean(state_count$n) #calculates the avg number of listings across states in this  
avg_count
```

```
[1] 194.5686
```

```
valid_states <- state_count %>%  
  filter(n >= avg_count) %>% #only includes states from the data set that have at least an avg  
  pull(state) #extracts the state name  
exp_states <- RentWIP %>%  
  filter(state %in% valid_states) %>% #filters out states that do not have at least the avg  
  group_by(state) %>%  
  summarize(avg_monthly_price = mean(monthly_price, na.rm = TRUE)) %>% #for each state, the  
  arrange(desc(avg_monthly_price)) %>% #orders avg monthly price from largest to smallest  
  slice_head(n = 25) #only show the top 25 most expensive states
```

```
ggplot(exp_states, aes(x = reorder(state, avg_monthly_price),
                        y = avg_monthly_price)) +
  geom_col(fill = "brown") +
  coord_flip() +
  labs(
    title = "Top 25 Most Expensive States to Rent",
    x = "State",
    y = "Average Monthly Rent ($)"
  ) +
  theme_classic()
```



```
city_count <- RentWIP %>% #same process from code chunk 11, but with cities instead of states
  filter(cityname != "null") %>%
  count(cityname)
avg_citycount <- mean(city_count$n)
avg_citycount
```

```
[1] 6.30432
```

```
valid_cities <- city_count %>% #creates a vector where only cities that have at least the avg
  filter(n >= avg_citycount) %>%
```

```

pull(cityname)
exp_cities <- RentWIP %>%
  filter(cityname %in% valid_cities) %>%
  group_by(cityname)

```

```

city_summary <- exp_cities %>% #generates a table with only cities that have at least the avg
  summarize(
    avg_price = mean(monthly_price, na.rm = TRUE),
    avg_sqft = mean(square_feet, na.rm = TRUE),
    listings = n()
  ) %>%
  arrange(desc(avg_price))
city_summary

```

```

# A tibble: 274 x 4
  cityname      avg_price avg_sqft listings
  <fct>         <dbl>    <dbl>    <int>
1 coronado      5512.    1797.      8
2 palo alto     4694.    1053.      9
3 beverly hills 4431      1149.      7
4 west hollywood 4419.    1132.     11
5 san francisco 3569.     614.    104
6 santa monica  3519.     965.     16
7 boston        3178.     949.     27
8 san jose      3165.    1367.     15
9 los angeles   3117.     941.    165
10 jersey city  3055.     877.     39
# i 264 more rows

```

```

RentWIP <- RentWIP %>%
  select(-fee, -id, -title, -body, -currency, -has_photo, -price, -price_display, -price_type)

```

```

avg_rent_pets <- RentWIP %>%
  group_by(pets_allowed) %>%
  summarize(
    avg_monthly_rent = mean(monthly_price, na.rm = TRUE),
    listings = n()
  )
avg_rent_pets

```

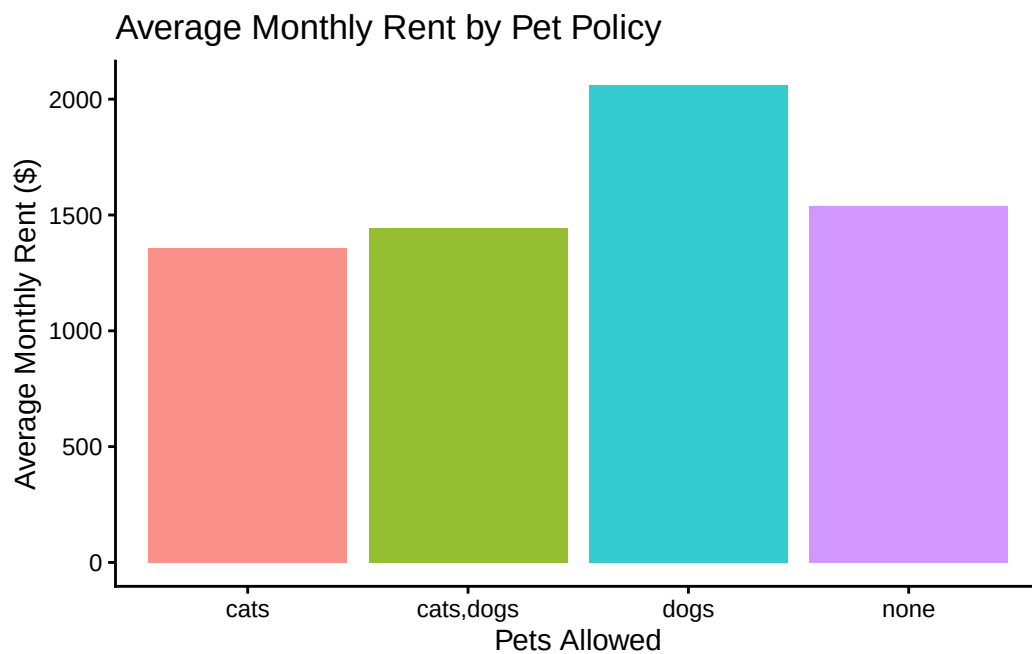
```

# A tibble: 4 x 3

```

	<code>pets_allowed</code>	<code>avg_monthly_rent</code>	<code>listings</code>
	<chr>	<dbl>	<int>
1	cats	1359.	485
2	cats,dogs	1444.	5228
3	dogs	2061.	124
4	none	1538.	4163

```
ggplot(avg_rent_pets, aes(x = pets_allowed, y = avg_monthly_rent, fill = pets_allowed)) +
  geom_col(alpha = 0.8) +
  labs(
    title = "Average Monthly Rent by Pet Policy",
    x = "Pets Allowed",
    y = "Average Monthly Rent ($)"
  ) +
  theme_classic() +
  theme(legend.position = "none")
```



```
write_csv(RentWIP, "RentWIP_cleaned.csv")
```